

Item Code Studio

User Handbook

Complete Guide to Managing Item Codes & ERPNext Synchronization

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TABLE OF CONTENTS

1. GETTING STARTED

- 1.1 What is Item Code Studio?
- 1.2 Logging In
- 1.3 Navigation

2. CREATE CODES (Code Generator)

- 2.1 Two Input Methods
- 2.2 Resolution Results
- 2.3 Editing & Fine-Tuning
- 2.4 Tax Template & HSN Code
- 2.5 Creating a New Group
- 2.6 Success — Code Issued!
- 2.7 Handling Duplicates

3. ITEM MASTER (The Working Master)

- 3.1 Overview
- 3.2 Filtering & Searching
- 3.3 Editing an Item
- 3.4 ERP-Only Items
- 3.5 Classifying ERP-Only Items
- 3.6 Pushing to ERPNext
- 3.7 Export to Excel

4. DICTIONARY (Taxonomy Manager)

- 4.1 Overview
- 4.2 Group Details
- 4.3 Managing Groups
- 4.4 Creating New Taxonomy
- 4.5 Sync from ERPNext ★ NEW

5. DECODER

6. ACTIVITY LOG

7. SETTINGS

8. QUICK REFERENCE

1. GETTING STARTED

1.1 What is Item Code Studio?

A web-based tool that generates mathematically structured 15-digit item codes. Every item in your inventory gets a unique code built from a hierarchy:

- **Head** (2 letters) — e.g., RM = Raw Materials, AO = Admin & Office, CA = Capital Assets
- **Sub-head** (2 letters) — e.g., BS = Battery Scrap, CH = Chemicals
- **Group** (3 digits) — e.g., 001 = Cobalt Hydroxide, 002 = Battery Pack
- **Spec 1-4** (2 digits each) — Size, Type, Chemistry, etc.
- **Vendor** (2 digits) — Supplier code

Example: **RMBS 001 02 02 01 01 00** = Raw Materials > Battery Scrap > Cobalt Hydroxide > specific specs

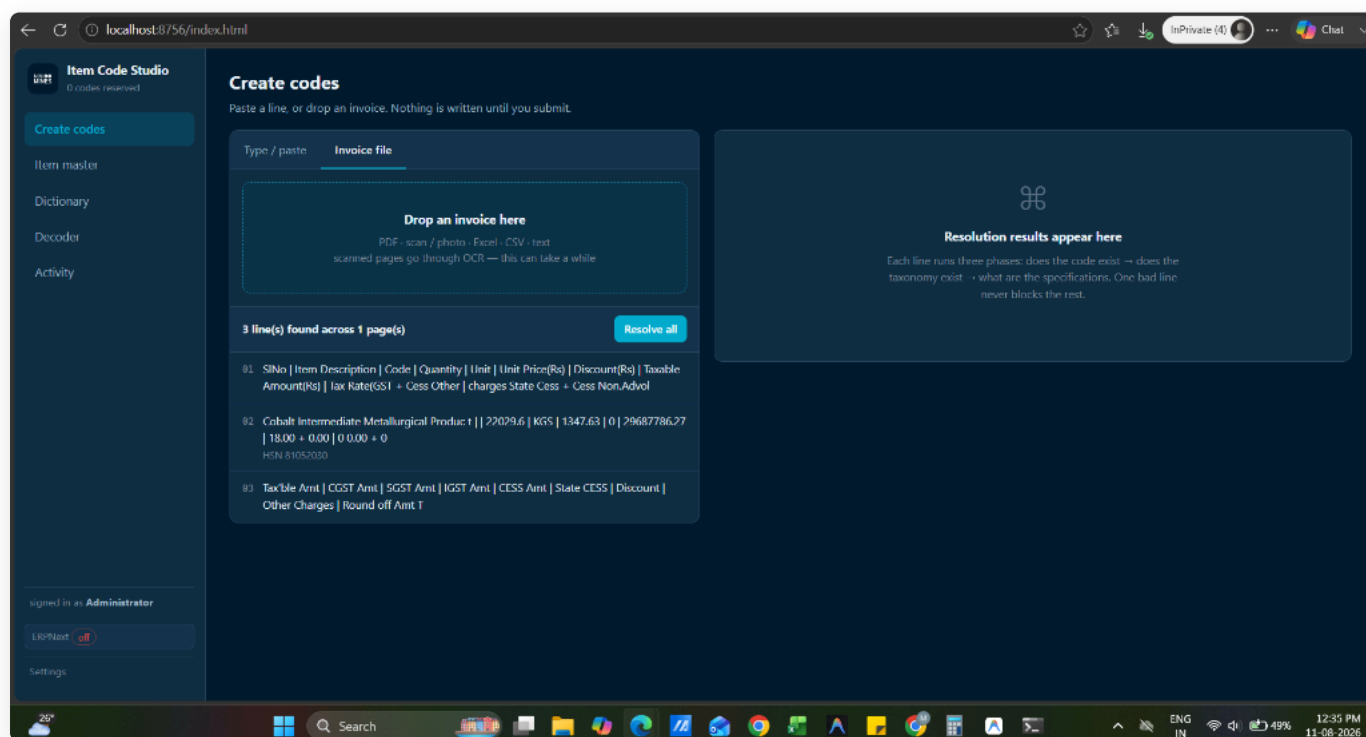
1.2 Logging In

Open the app URL in your browser. Enter your username and password. Contact your administrator for credentials.

1.3 Navigation

The left sidebar has 5 main tabs:

- **Create codes** — Generate new item codes
- **Item master** — View/edit all items
- **Dictionary** — Manage taxonomy (Heads, Sub-heads, Groups)
- **Decoder** — Reverse-lookup any code
- **Activity** — Audit trail of all changes



The main interface showing the sidebar navigation and Create Codes workspace

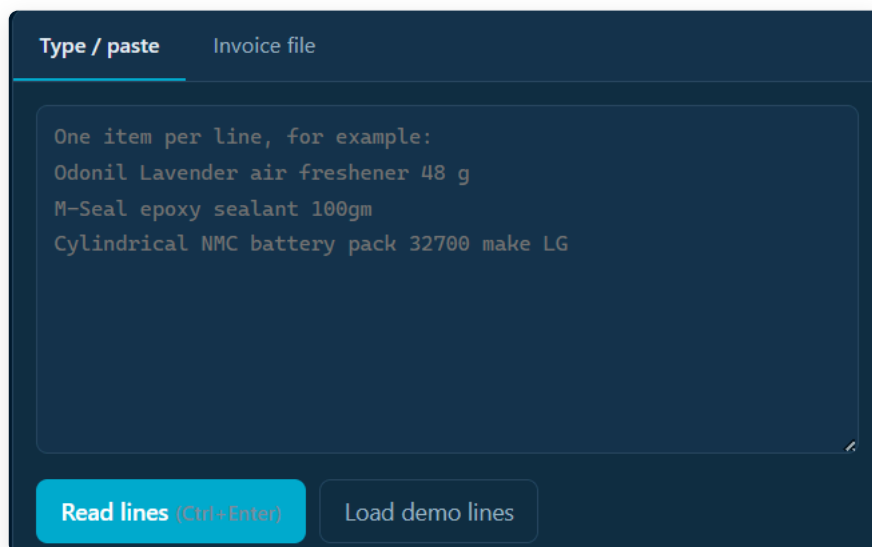
2. CREATE CODES (Code Generator)

This is the primary workspace where you mint new item codes.

2.1 Two Input Methods

Method A: Type / Paste

Manually type or paste item descriptions, one per line.



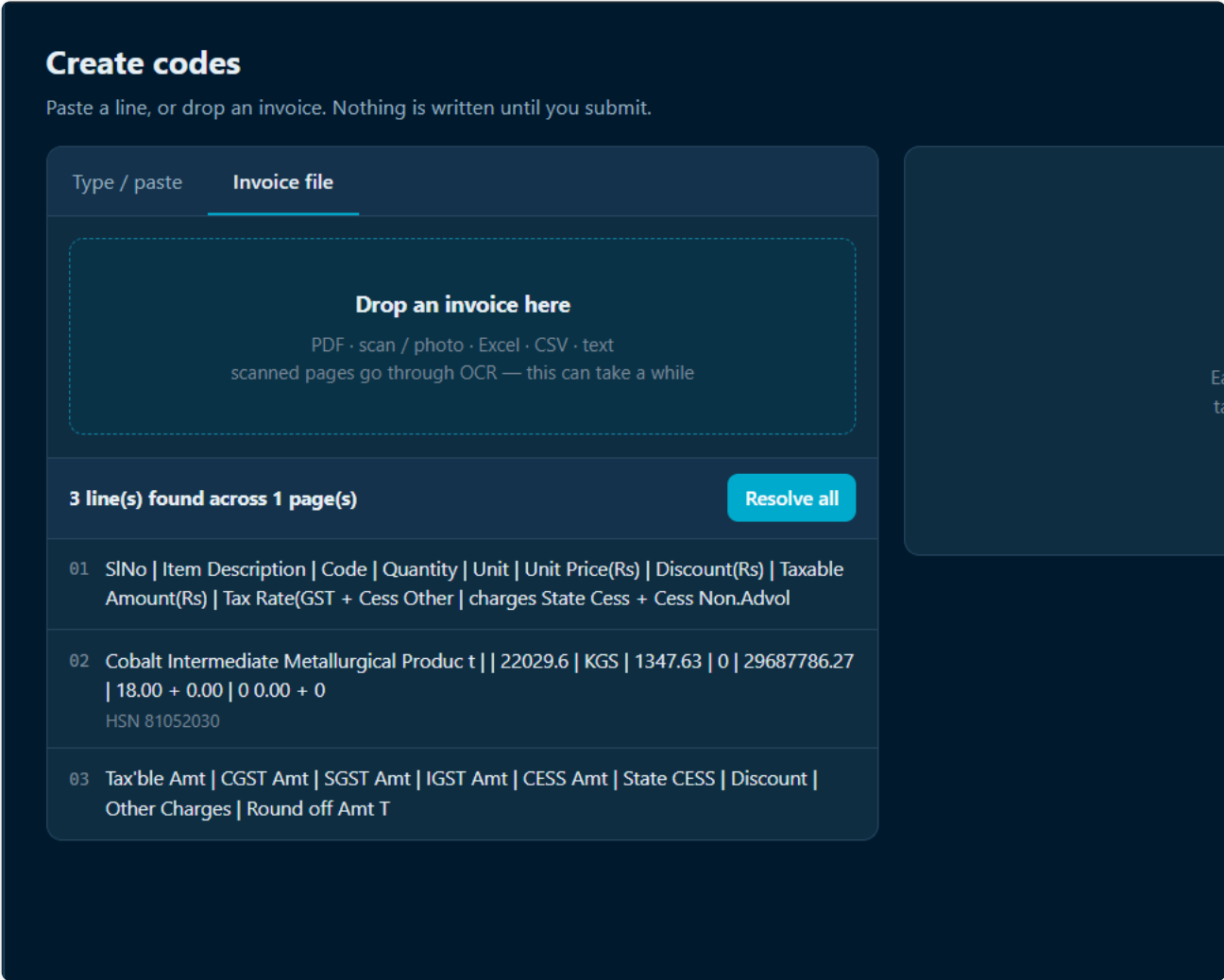
The screenshot shows a dark-themed interface with two tabs at the top: 'Type / paste' and 'Invoice file'. The 'Type / paste' tab is active and contains a large text input area. Inside the text area, there is a prompt 'One item per line, for example:' followed by four example lines of item descriptions: 'Odonil Lavender air freshener 48 g', 'M-Seal epoxy sealant 100gm', and 'Cylindrical NMC battery pack 32700 make LG'. Below the text area, there are two buttons: a blue button labeled 'Read lines (Ctrl+Enter)' and a grey button labeled 'Load demo lines'.

The Type/Paste input area with example items

- 1 Click the **Type / paste** sub-tab
- 2 Enter item names, one per line (e.g., "Cobalt Hydroxide 500g", "Laptop Acer Aspire 5")
- 3 Click **Read lines** to process them
- 4 The AI engine will classify each item automatically

Method B: Invoice File (OCR)

Upload a purchase invoice PDF. The built-in OCR engine extracts line items automatically.

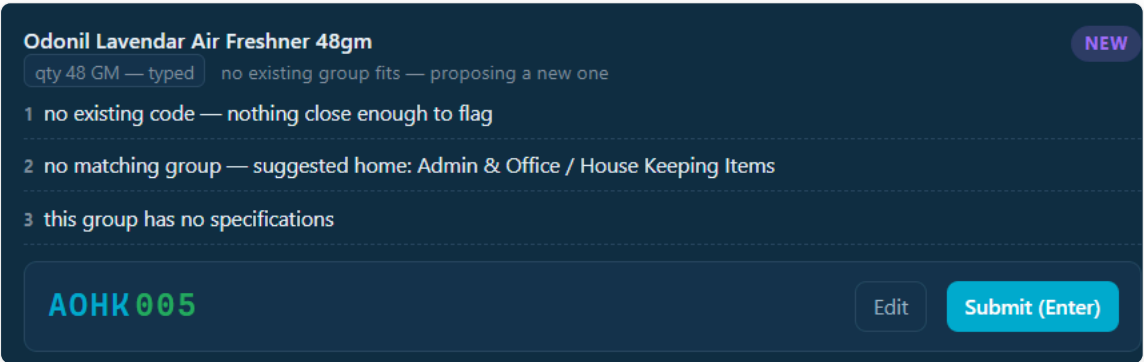


OCR-parsed invoice lines ready for code generation

- 1 Click the **Invoice file** sub-tab
- 2 Upload a PDF invoice
- 3 The OCR engine extracts item descriptions, quantities, HSN codes
- 4 Click **Resolve all** to auto-classify every line

2.2 Resolution Results

After processing, each item shows a resolution card on the right panel:



A NEW item auto-matched to Admin & Office > House Keeping Items

The resolution shows:

- **Phase 1 — Exists?** checks if the item already exists

- **Phase 2 — Taxonomy** auto-classifies into Head > Sub-head > Group
- **Phase 3 — Specs** identifies specification slots

2.3 Editing & Fine-Tuning

If the AI classification needs adjustment, click the item to open the editing form:

Civil Works Providing and constructing Discharge Tank Tank of size 2.60x2.50x3.90m - 2 Nos Item includes Earth work excavation, Bed concrete, 9" thick Burnt Brick masonry, Inside Double Plastering (rough plastering with mesh & smooth finishing). The cost includes labour, material, curing, mixing, etc., complete 2.60x2.50x3.90m - 2 Nos

qty 50700 NOS · via OCR — from invoice

proposed code

Head

Raw Materials (RM)

Sub-head

Battery Scrap (BS)

Group

Battery Pack (001)

Chemistry

cobalt22% · new

02

Type

che · new

02

Spec 3

0 · new

01

Spec 4

0 · new

01

Vendor/Brand

Unorg · 00

00

RMBS0010202010100

Cancel (Esc)

Submit (Enter)

■ head + sub-head

■ group

■ specification

■ 00 = not applicable

■ vendor

The editing form with Head, Sub-head, Group, and Spec dropdowns. The color-coded item code builds live at the bottom.

- 1 Select the correct**Head**from the dropdown
- 2 Select the correct**Sub-head**
- 3 Select or create a**Group**
- 4 Fill in**Spec 1-4**values from dropdowns (or type a new value)
- 5 The 15-digit code builds live at the bottom, color-coded by segment
- 6 Click**Submit**to mint the code

Item Code Studio
2,694 codes reserved

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in as Administrator

ext · free

Create codes

Paste a line, or drop an invoice. Nothing is written until you submit.

Type / paste

Invoice file

testcodeerp

Read lines (CSV/JSON)

Load demo lines

1 line from typed text

Resolve all

01 testcodeerp

testcodeerp

proposed code

Head

Raw Materials (RM)

Sub-head

Chemicals (CH)

Group

odonil-test (013)

odonil

2

--

airfresher jasmine

3

--

Spec 3

— none —

--

Spec 4

2

--

Vendor

— none —

--

Tax Template *

GST 18% - MC

--

HSN Code (if required)

Enter HSN Code

--

RMCH01301010001

Cancel (Esc)

Submitting...

■ head + sub-head

■ group

■ specification

■ 00 = not applicable

■ vendor

2.4 Tax Template & HSN Code

When editing an item, you can also set:

- **Tax Template** — GST rate (e.g., GST 18% - MC)
- **HSN Code** — Harmonized System code for the item

Head

Consumables (CO)

Sub-head

Packaging (PA)

Group

Pallets (005)

Spec 1

— none —

Spec 2

45

Spec 3

11

Spec 4

— none —

Vendor

— none —

Tax Template *

GST 18% - MC

HSN Code (if required)

528525

COPA005

Cancel (Esc)

Submit (Enter)

■ head + sub-head ■ group ■ specification ■ 00 = not applicable ■ vendor

COPA005 is already issued - adjust a spec value or pick the existing item.

Tax Template and HSN Code fields in the editing form

2.5 Creating a New Group

If no existing group fits, create one inline:

Type: U+2 panel bag with 4-corner loops Top: Fully OPEN Bottom: FLAT 90x90x110 cm 140 GSM;
SWL: 1 Ton Fabric Color: White Liner: NO; Lamination: No; Printing: No; Document pouch: No; Pallet:
No; Lead time: 1 day || 250 | 240.00 | bag | 60,000.00

HSN 63053200 — from invoice proposed code

Head **Consumables (CO)** Sub-head **Packaging (PA)** Group **+ create a new group**

New group

UoM

Spec labels

Vendor label

— not enough information —

■ head + sub-head ■ group ■ specification ■ 00 = not applicable ■ vendor

choose a head, sub-head and group

Creating a new group with name, UoM, spec labels, and vendor label

- 1 Select "Create a new group" option
- 2 Enter the group name
- 3 Set the Unit of Measurement
- 4 Define spec label names (e.g., Type, Size, Chemistry)
- 5 Click **Apply** to create and assign

2.6 Success — Code Issued!

Once submitted, you see a green confirmation:

laptop acer aspire 5 **NEW**

proposed code

1 no existing code nearest: Acer Laptop (78%)

2 Laptop matched 65%

3

CAFF007

■ head + sub-head ■ group ■ specification ■ 00 = not applicable ■ vendor

Code successfully issued! The system also learns the wording for future auto-matching.

TIP

The system learns from every code you create. Next time you type a similar item name, it will auto-match to the correct group!

2.7 Handling Duplicates

If an item already exists, the system warns you:

SHAFT BATTERIES

ALREADY EXISTS

phase 1 stopped here — this item already has a code

RMBS0010202

 SHAFT BATTERIES

item master · identical after normalisation · 100%

Copy existing code

 Or continue editing below to create a new one

Head

Raw Materials (RM)

Sub-head

Battery Scrap (BS)

Group

Battery Pack (001)

Chemistry

NMC

Type

CYLINDRICAL

Spec 3

red

Spec 4

— none —

Vendor/Brand

— none —

Tax Template *

GST 18% - MC

HSN Code (if required)

Enter HSN Code

Two items resolved: one already exists (100% match), one is new

WARNING

Always check the 'already exists' warning before creating duplicate codes!

3. ITEM MASTER (The Working Master)

The central repository of ALL items — both locally generated and ERP-only.

3.1 Overview

Item master

The working master. Edits here change fields, never codes.

Export to Excel

search code, name or description...

any head

any sub-head

any group

any status

only codes that no longer decode

13 items • 2689 ERP-only

CODE	ITEM NAME	HEAD / SUB-HEAD	GROUP	SPECIFICATIONS	VENDOR	UOM	HSN	STATUS	VER.	
RMCH01301	WFHTEST	Raw Materials Chemicals	odonil-test	2	—			in ERPNext	2	edit
RMCH0130101	Odonil Lavendar Air Freshner 48gm	Raw Materials Chemicals	odonil-test	2 · 3	—	GM		in ERPNext	1	edit
RMCH01301010001	testcodeerp	Raw Materials Chemicals	odonil-test	2 · 3 · 2	—			confirmed	1	edit
001	P - Hub Stairs and Railing work	—	Consumable	—	—	Kg	—	ERP only	—	view in ERP
00301	Argon Gas (Spectra)	—	Argon Gas (Spectra)	—	—	Nos	—	ERP only	—	view in ERP
00501	Transportation Charges	—	Transportations	—	—	Nos	—	ERP only	—	view in ERP
005030602	Lithium IOM	—	Cell	—	—	Kg	—	ERP only	—	view in ERP
0050001	Eira Estensioischer Duffillies	—	Eira Estensioischer Duffillies	—	—	Misc	—	ERP only	—	view in ERP

Push to ERP

prev

page 1 of 1

next

Item Master showing a mix of confirmed, in ERPNext, and ERP-only items with Push to ERP button

The grid shows:

- **CODE** — The 15-digit item code (or ERP code for ERP-only items)
- **ITEM NAME** — Description
- **HEAD / SUB-HEAD** — Taxonomy classification
- **GROUP** — Product group
- **SPECIFICATIONS** — Spec values (shown as slot numbers like 2·3·2)
- **VENDOR** — Supplier
- **UOM** — Unit of measurement
- **HSN** — HSN/SAC code
- **STATUS** — confirmed / in ERPNext / ERP only
- **VER.** — Version number

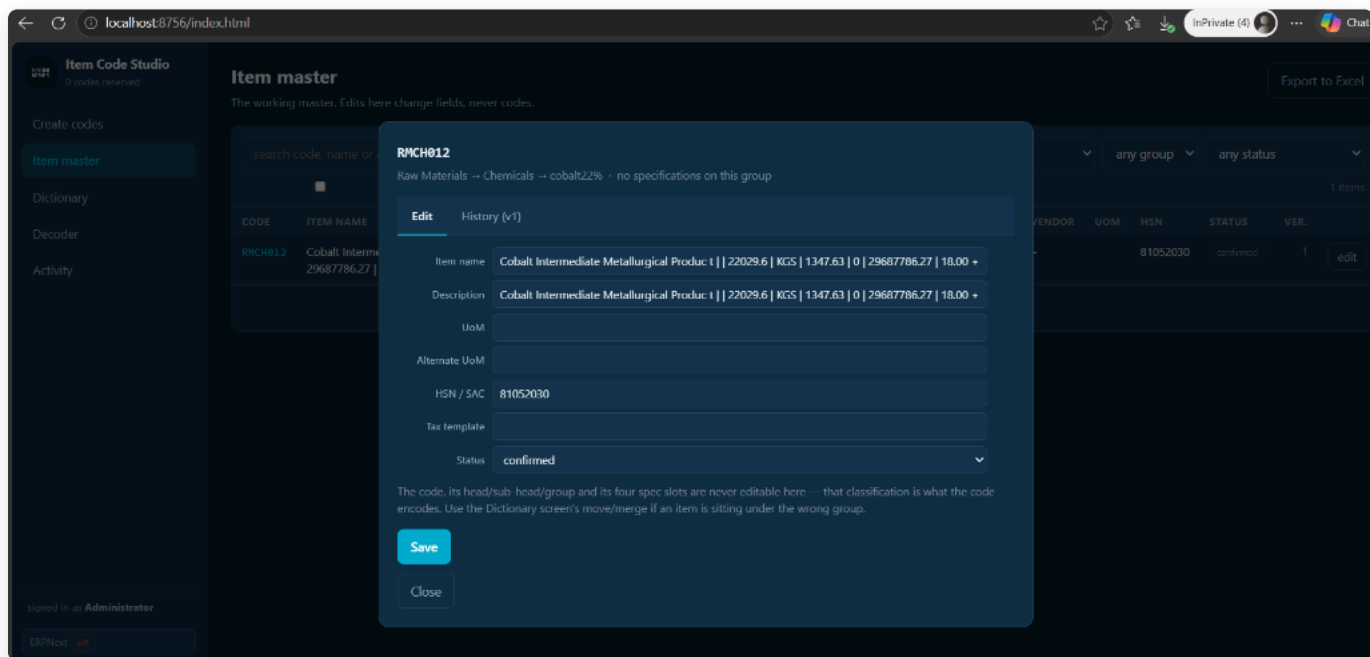
3.2 Filtering & Searching

Use the toolbar at the top:

- **Search box** — Search by code, name, or description
- **Head filter** — Filter by Head category
- **Sub-head filter** — Filter by Sub-head
- **Group filter** — Filter by Group
- **Status filter** — Filter by status (confirmed, in ERPNext, ERP only)

3.3 Editing an Item

Click the **edit** button on any locally-created item to modify its details:



Edit modal for modifying item name, description, UoM, HSN, tax template, and status

3.4 ERP-Only Items

Items created directly in ERPNext appear with an orange **ERP only** badge:

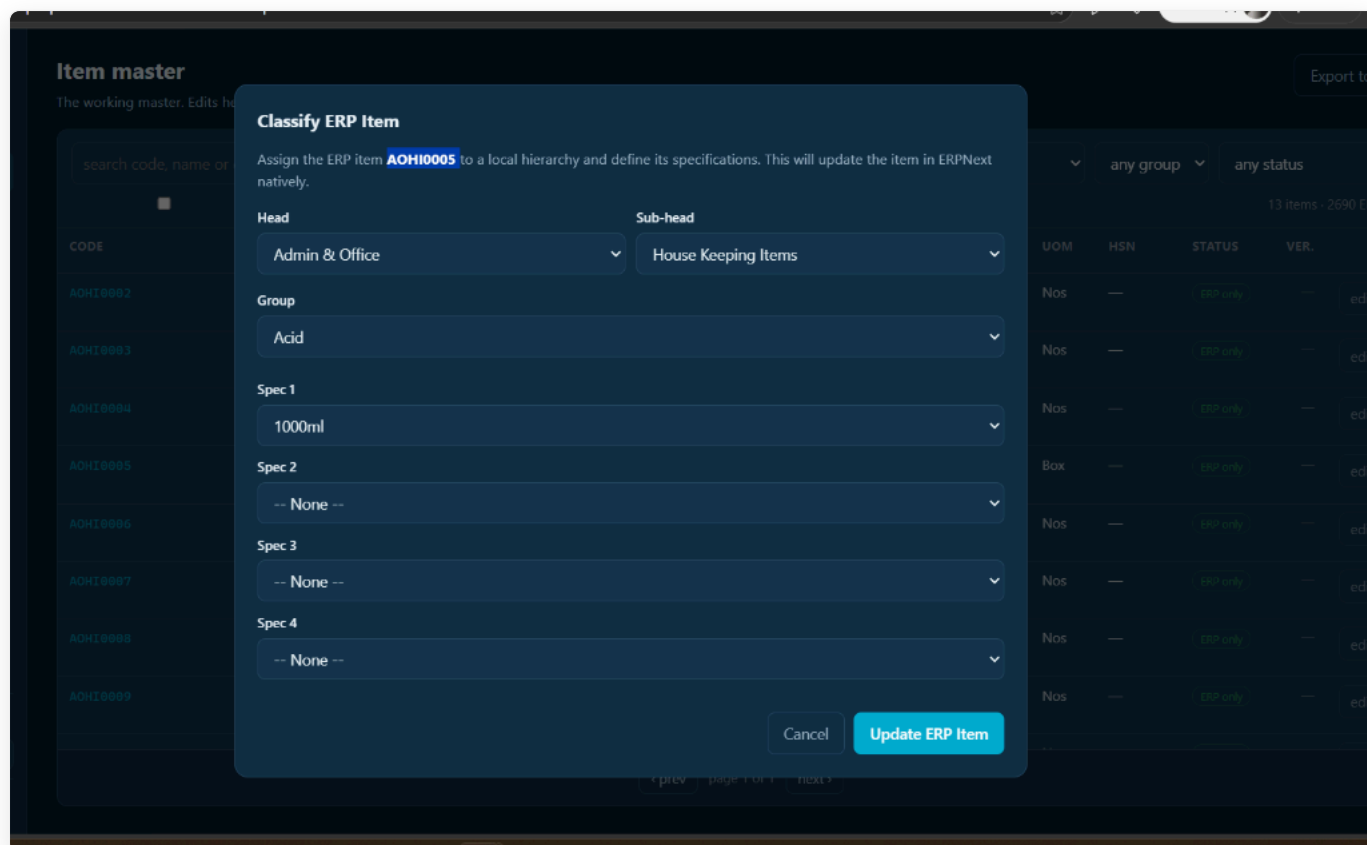
CODE	ITEM NAME	HEAD / SUB-HEAD	GROUP	SPECIFICATIONS	VENDOR	UOM	HSN	STATUS	VER.
ADHK0040101	Bucket - Standard SS	Admin & Office House Keeping	Bucket	—	—	Nos	—	ERP only	view in ERP
ADHK004020001	Bucket - PL Turbo 25 L	Admin & Office House Keeping	Bucket	—	—	Nos	—	ERP only	view in ERP
ADHK004030002	Bucket - Samruddhi 10 L	Admin & Office House Keeping	Bucket	—	—	Nos	—	ERP only	view in ERP
ADHK00501	Cleaning Brush - Iron	Admin & Office House Keeping	Cleaning Brush	—	—	Nos	—	ERP only	view in ERP
ADHK0050201	Cleaning Brush - Handle Wooden	Admin & Office House Keeping	Cleaning Brush	—	—	Nos	—	ERP only	view in ERP
ADHK00503	Cleaning Brush - Ceiling	Admin & Office House Keeping	Cleaning Brush	—	—	Units	—	ERP only	view in ERP
ADHK00601	Cleaning Cloth - Checks	Admin & Office House Keeping	Cleaning Cloth	—	—	Nos	—	ERP only	view in ERP
ADHK007	Covers	Admin & Office House Keeping	Covers	—	—	Nos	—	ERP only	view in ERP
ANUS00001	Rotamant Drivler - Wheel	Admin & Office	Rotamant Drivler	—	—	Kn	—	ERP only	view in ERP

ERP-only items from Admin & Office > House Keeping showing Buckets, Cleaning Brushes, etc.

These items exist in ERPNext but don't have a structured 15-digit code yet.

3.5 Classifying ERP-Only Items

To bring an ERP-only item into your structured taxonomy:



Classify ERP Item modal — assigning Head, Sub-head, Group, and Specifications to ERP item AOH10005

- 1 Find the ERP-only item in the master grid
- 2 Click the **edit** button on its row
- 3 The **Classify ERP Item** modal opens
- 4 Select the correct **Head** (e.g., Admin & Office)
- 5 Select the correct **Sub-head** (e.g., House Keeping Items)
- 6 Select the correct **Group** (e.g., Acid)
- 7 Set **Spec 1-4** values from the dropdowns
- 8 Click **Update ERP Item**

The item's metadata is updated directly in ERPNext without changing its original code.

IMPORTANT

This does NOT rename the item code. It only assigns structured metadata (Head, Sub-head, Group, Specs) to the existing ERP item.

3.6 Pushing to ERPNext

Items with status 'confirmed' can be pushed to ERPNext:

- Click the **Push to ERP** button on the item row
- The item and its metadata are synced to your ERPNext instance

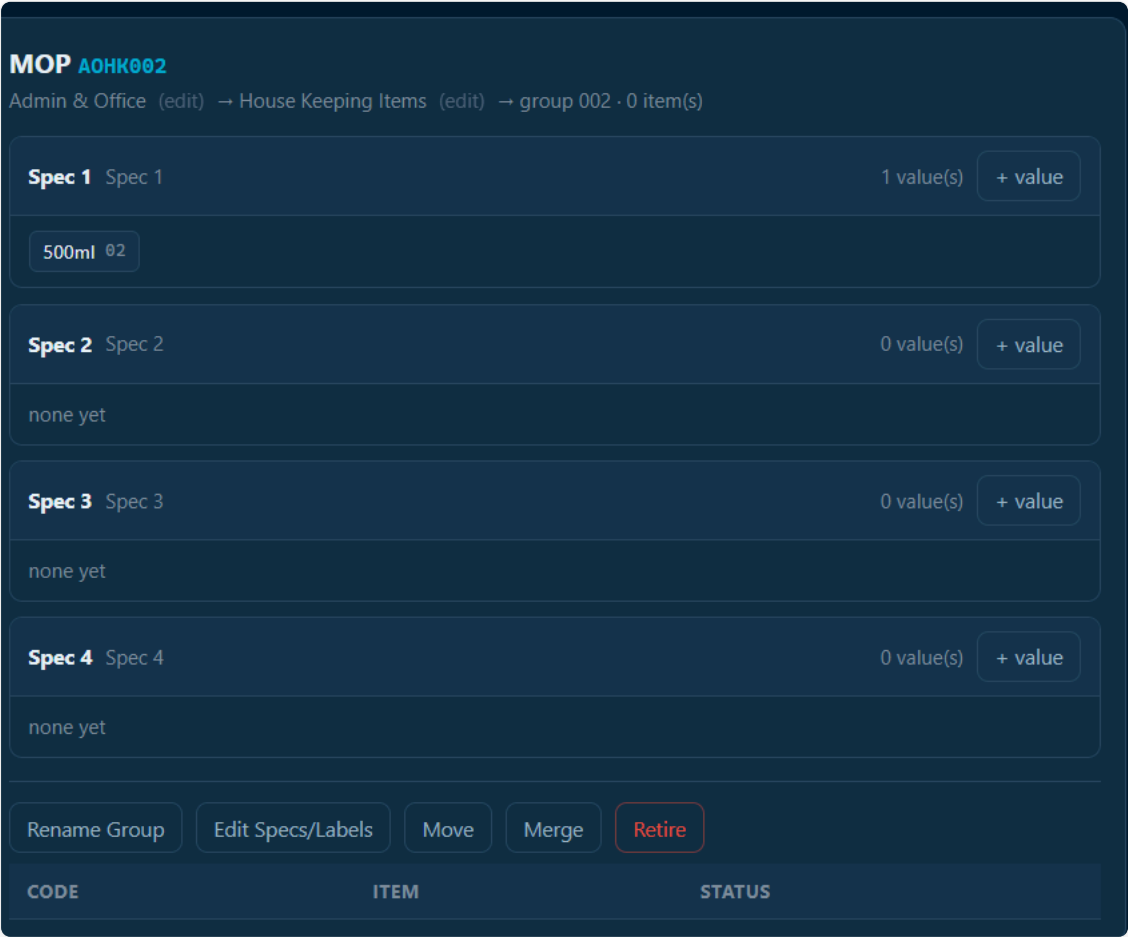
3.7 Export to Excel

Click the **Export to Excel** button in the top right to download the entire master as a spreadsheet.

4. DICTIONARY (Taxonomy Manager)

The Dictionary controls the DNA of your codes — all Heads, Sub-heads, Groups, and their specification slots.

4.1 Overview



Dictionary showing group detail for MOP (AOHK002) with spec slots, and action buttons

The left panel lists all groups with their prefix codes. Click any group to see its details on the right.

4.2 Group Details

Each group shows:

- **Prefix code** (e.g., AOHK002)
- **Head / Sub-head** path
- **Spec slots** (1-4) with their labels and defined values
- **Action buttons:** Rename, Edit Specs/Labels, Move, Merge, Retire

4.3 Managing Groups

Rename a Group

Click **Rename Group** → Enter the new name → The old name becomes a search alias.

Edit Specs & Labels

Click **Edit Specs/Labels** → Modify spec slot names or add new specification values.

Move a Group

Click **Move** → Select a different Sub-head → Preview impact → Confirm.

NOTE

Moving a group recalculates all item codes under it. Active ERPNext items are never broken.

Merge Groups

Click **Merge** → Select the target group → All items from the source move to the target.

Retire a Group

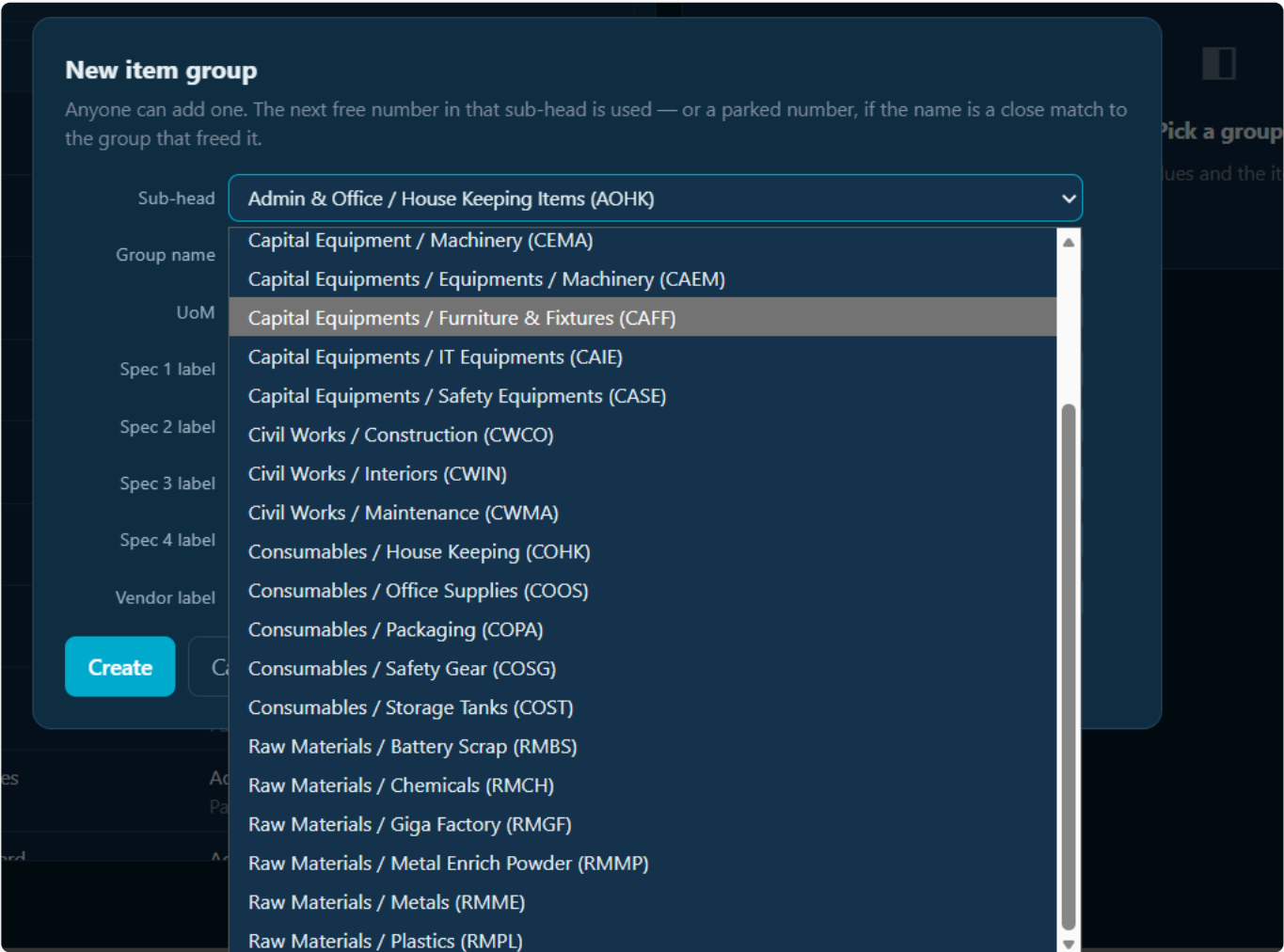
Click **Retire** → The group is deactivated and its 3-digit code is permanently parked.

WARNING

Retired group codes are never reused to prevent confusion with historical records.

4.4 Creating New Taxonomy

Click **New group** to open the creation modal:



New hierarchy modal showing all available Head/Sub-head combinations

You can create:

- A new **Group** under an existing Sub-head
- A new **Sub-head & Group** under an existing Head
- A new **Head, Sub-head & Group** from scratch

4.5 Sync from ERPNext ★ NEW

If new Item Groups are created directly in ERPNext, use this feature to import them locally.

- 1 Click **Sync from ERPNext** button (top right of Dictionary page)
- 2 The system analyzes ERPNext and shows missing Heads, Sub-heads, and Groups
- 3 Click **Import** next to any missing item
- 4 The system auto-generates the 2-letter abbreviation code
- 5 The taxonomy is now available in the generator!

IMPORTANT

Always import parent Heads before their child Sub-heads, and Sub-heads before their Groups.

5. DECODER

Reverse-engineer any 15-digit item code to see what it means.

- 1 Navigate to the **Decodertab**
- 2 Enter any valid item code (e.g., RMBS0010202010100)
- 3 The system instantly breaks down every segment:
 - Head → Raw Materials
 - Sub-head → Battery Scrap
 - Group → Cobalt Hydroxide
 - Spec 1 → NMC (02)
 - Spec 2 → Cylindrical (02)
 - etc.

TIP

Use the Decoder to verify codes before sharing them with suppliers or entering them in purchase orders.

6. ACTIVITY LOG

A complete audit trail of every action taken in the system.

What's logged:

- Item code creation (who, when, what)
- Taxonomy changes (rename, move, merge, retire)
- ERP sync events
- Item edits and version history

Click any event to see a **visual diff** showing exactly what changed (before vs. after).

TIP

Use the Activity log to track who made changes and when. This is essential for audit compliance.

7. SETTINGS

Access Settings from the bottom of the sidebar.

Configurable options:

- **ERPNext URL** — Your ERPNext instance URL
- **API Key & Secret** — Authentication credentials for ERPNext API
- **LLM Provider** — AI engine settings (OpenAI, etc.)
- **Auto-push** — Automatically push new codes to ERPNext

WARNING

Keep your API credentials secure. Only administrators should access the Settings page.

8. QUICK REFERENCE

Task	Where	Steps
Create a new item code	Create Codes → Type/Paste	Enter name → Resolve → Edit specs → Submit
Import from invoice	Create Codes → Invoice File	Upload PDF → Resolve All → Review → Submit
Edit existing item	Item Master → edit button	Modify fields → Save
Classify ERP item	Item Master → edit (ERP only)	Set Head/Sub/Group/Specs → Update
Push to ERPNext	Item Master → Push to ERP	Click button → Confirm
Add new group	Dictionary → New group	Set hierarchy → Define specs → Create
Sync ERP taxonomy	Dictionary → Sync from ERPNext	Review missing → Import → Done
Decode a code	Decoder	Enter code → View breakdown
View audit trail	Activity	Browse events → Click for diff

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